

A Preliminary Approach to Investigate Recent Increases in New York Tornado Activity: A Wavelet Analysis and Forecasting Approach using R



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Abstract:

When considering tornadoes, one's first thought likely veers towards the infamous "Tornado Alley" that is a region in the Midwest where tornadoes are notoriously frequent and severe. New York's tornado frequency, in contrast, is nowhere near as high. Despite this, historical records show an increase in tornado events within the NY State. This raised questions about the existence of a pattern tied to these events and subsequent geohazards intensity. Utilizing tornado data gathered from MPNNow's tornado database, we aimed to both investigate a potential pattern and forecast future tornado rates. Wavelet analysis detects frequency variations over time and after applying the Kolmogorov-Zurbenko (KZ) filter, which excludes high-frequency components, and reveals annual and multi-annual cycles. We modelled our data using the Autoregressive Integrated Moving Average (ARIMA) function, allowing for a general forecast of potential future tornado frequencies. Geologically, tornadoes are known to affect multiple processes, including but not limited to soil erosion, sediment redistribution, and mass wasting. The existence of a multi-annual tornado-driving cycle would provide new insights in our understanding of tornadoes, which could prove useful in preparation and damage mitigation efforts.

Introduction:

Tornadoes are most commonly associated with "Tornado Alley" in the Midwest, a region that experiences a high frequency of tornadoes. In contrast, New York State has not traditionally been considered a hotspot for tornado activity. However, recent studies have shown a notable rise in tornado occurrences within the state, prompting questions about whether cyclical or trend-related phenomena might be driving these events. Tornadoes can have significant impacts on various geological processes, including soil erosion, sediment redistribution, and mass wasting, which makes understanding these trends crucial for improving risk management and preparedness. To explore this, we analyzed historical tornado data from MPNNow's database, supplemented by official reports from NOAA. We employed wavelet analysis to detect frequency variations over time, while using the Kolmogorov-Zurbenko (KZ) filter to remove low-frequency components, such as annual cycles, from the data. Additionally, we applied Fourier terms and ARIMA modeling to capture seasonal patterns and help forecast future tornado activity in the region.

Our study involved the pursuit of three distinct goals, which were:

1. To either determine or disprove the existence of a multi-annual tornado-driving cycle.
2. To, assuming the existence of the former, forecast future tornado frequencies.
3. To discuss the geological and environmental effects of increased tornado activity.

Image Description: Breezy Point, Queens, NY (2012)
(<https://www.cnn.com/2012/09/08/us/northeast-severe-weather/index.html>)

Materials & Methods:

- ❖ While prepping the data we collected it from MPNNow.com which utilizes the data collected from NOAA(National Oceanic and Atmospheric Administration). It was converted into a binary system of 1's and 0's a 1 meaning a tornado had occurred that day and 0 meaning that a tornado hadn't occurred. (Figure 1A)
- ❖ The 30 day rolling cumulative sum in the sum taken during any 30 day window. This was then smoothed out utilizing the KZ filter, which averages the amount of tornado occurrences over a given 365 days to eliminate the seasonal fluctuations. (Figure 1B)
- ❖ The wavelet function utilized the KZ moving average and was used to identify any possible dominant periodicities. (Figure 1C)
- ❖ The forecasting ARIMA model was utilized to give us our prediction. In order to get the most accurate data possible. (Figure 1D)
- ❖ We further filtered or smoothed our data by increasing the range that it created its moving average to account for the known changes that El Niño or La Niña can have on the atmospheric conditions so knowing that it is a 2-7 period we increased the range for the KZ function to 7 years which removed the effect that this Oscillation could have on our data.

Results:

- ❖ An analysis of tornado events in New York from May 6, 1952, to September 10, 2024, reveals a long-term upward trend in tornado frequency, with notable peaks around 2023 and 2024. A recurring cycle of approximately 13 to 17 years suggests tornado frequency follows a predictable pattern rather than occurring randomly. Wavelet analysis confirms this periodicity, showing strong 13 to 17-year cycles with high-power intervals indicating periods of increased activity. Auto ARIMA forecasting predicts a likely decline in tornado activity through 2027.

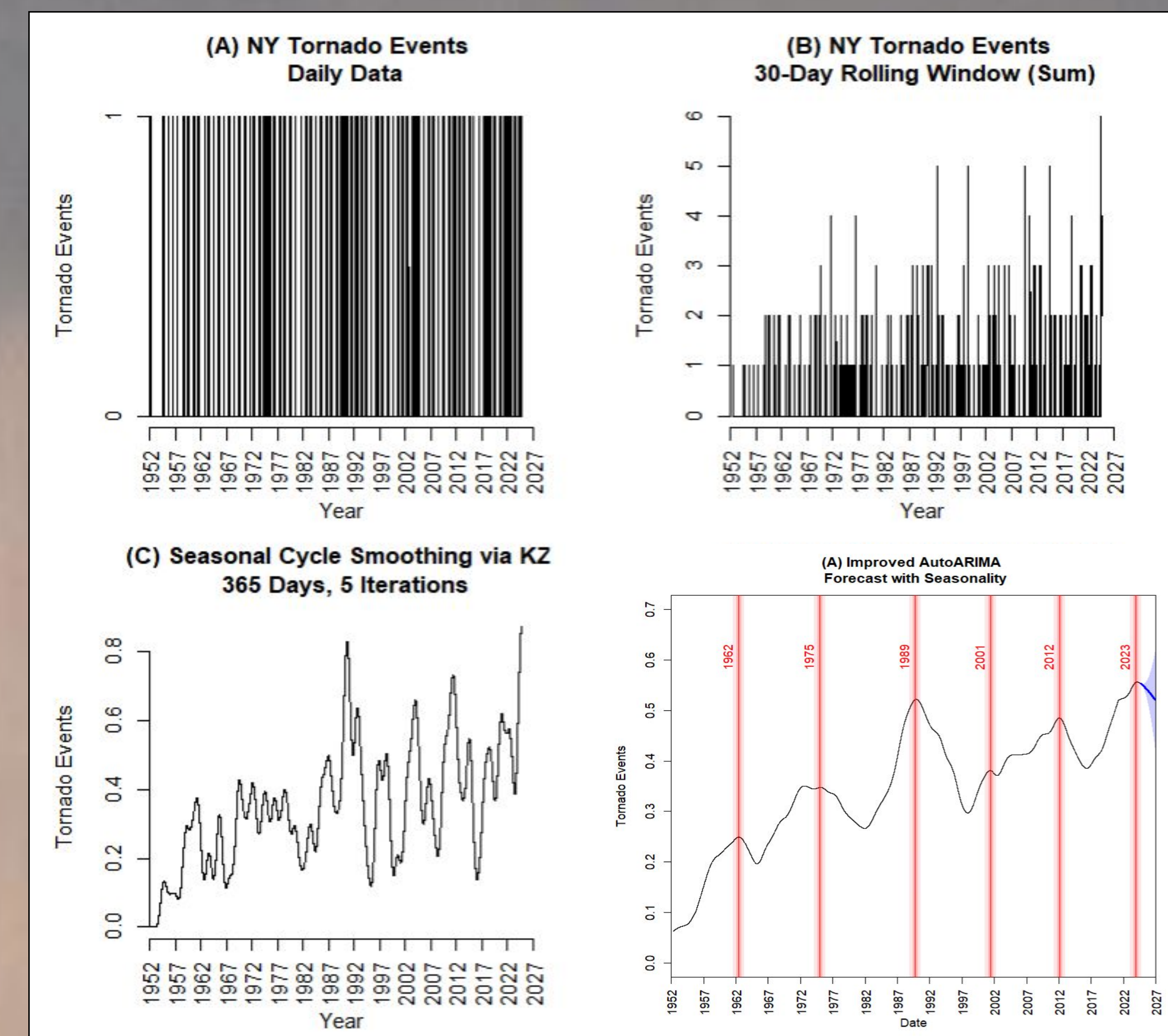


Figure 1: Demonstration of the 4 stages of data processing including prediction

Discussion/Conclusion:

- ❖ Our analysis reveals a recurring 13–17-year cycle in tornado activity, along with a long-term upward trend in both minima and maxima. Shorter decadal modulations, likely driven by interannual climate variability such as ENSO (2–7 years), contribute to localized peaks and valleys within the broader periodic pattern. Although La Niña's influence is weaker in the Northeast, our findings suggest a possible correlation, as tornado activity aligns with both La Niña and El Niño phases.
- ❖ Beyond meteorology, tornadoes have notable environmental impacts in areas like Long Island, where high winds accelerate erosion of nutrient-rich glacial till, potentially leading to nutrient loss and slope instability. Tornado-induced rainfall and storms can worsen flooding, while water-crossing tornadoes may stir sediments, raising ecological concerns.
- ❖ ARIMA modeling forecasts a decline in tornado frequency through 2027 following the recent peak. As shown in Fig. 2A, the model predicts continued cyclical activity, reinforcing the 13–17-year periodicity and overall increasing trend, with the shaded region indicating a 95% confidence interval.

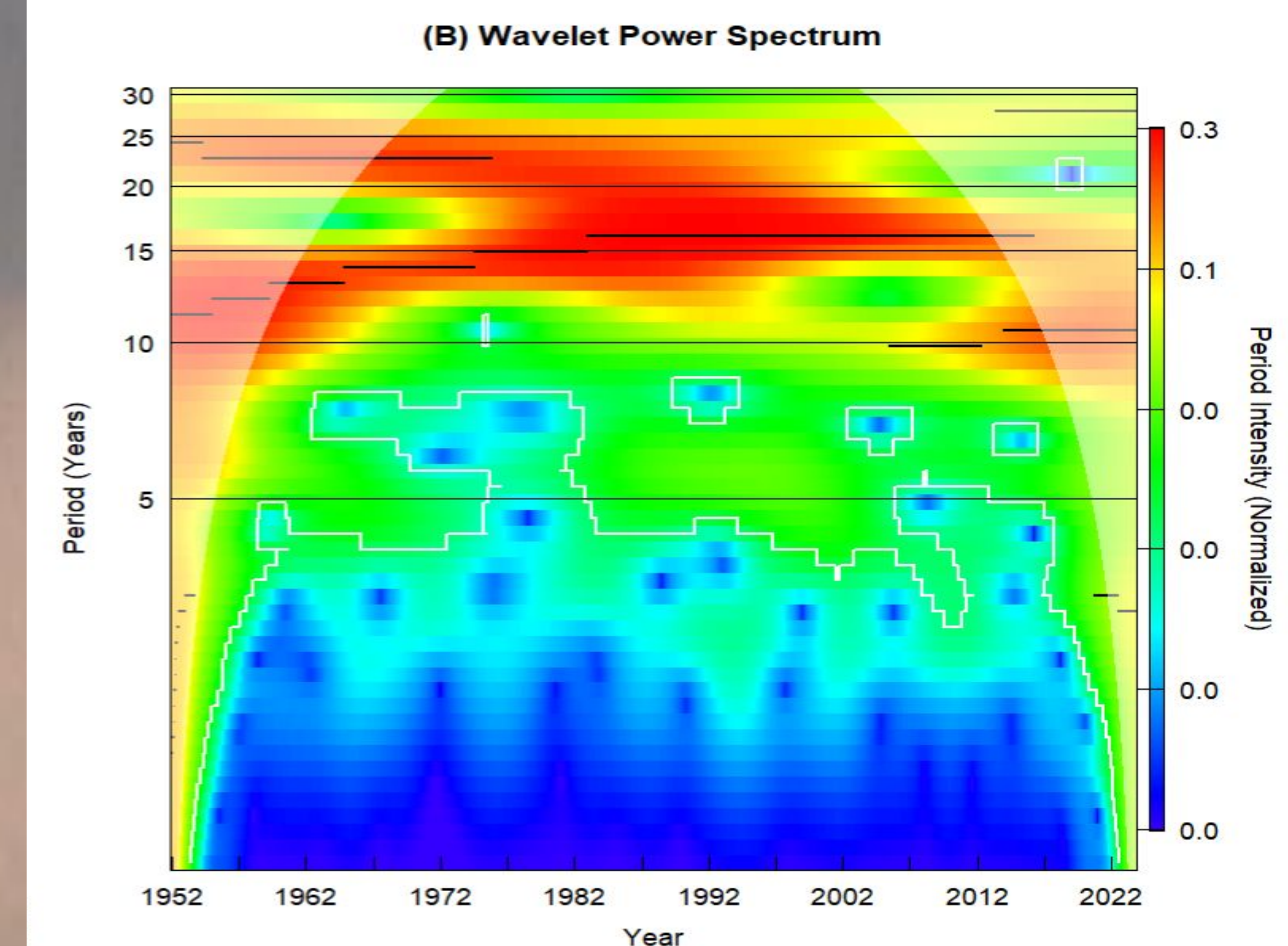


Figure 2: Wavelet Power Spectrum of New York Tornado Events.

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Credit Authorship Contribution Statement:

- ❖ **Dylan Hubbs** : Coding, Figure Construction, Formatting, Captions, General Editing
- ❖ **Xavier Gonzalez** : Discussion, Conclusion General Editing, coding
- ❖ **Adalia Rezki** : Abstract, Introduction, General Editing.
- ❖ **Kevin Tecusan** : Abstract, Results, General Editing
- ❖ **Antonios Marsellos** : Supervision, coding